

DESIGN AND IMPLEMENTATION OF A RELATIONAL UNIVERSITY LIBRARY MANAGEMENT SYSTEM



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- Course: Database Systems (CSC1204)

TRANSITIONING FROM MANUAL TO DIGITAL SYSTEMS



- The Problem: Current paper-based methods cause difficulty in tracking books, members, and late returns.
- The Solution: A modern relational database using MySQL to improve speed and data safety.
- Security: Mandatory Librarian passwords to prevent unauthorized record changes.

LOGICAL DESIGN

CORE ENTITIES OF THE LIBRARY SYSTEM

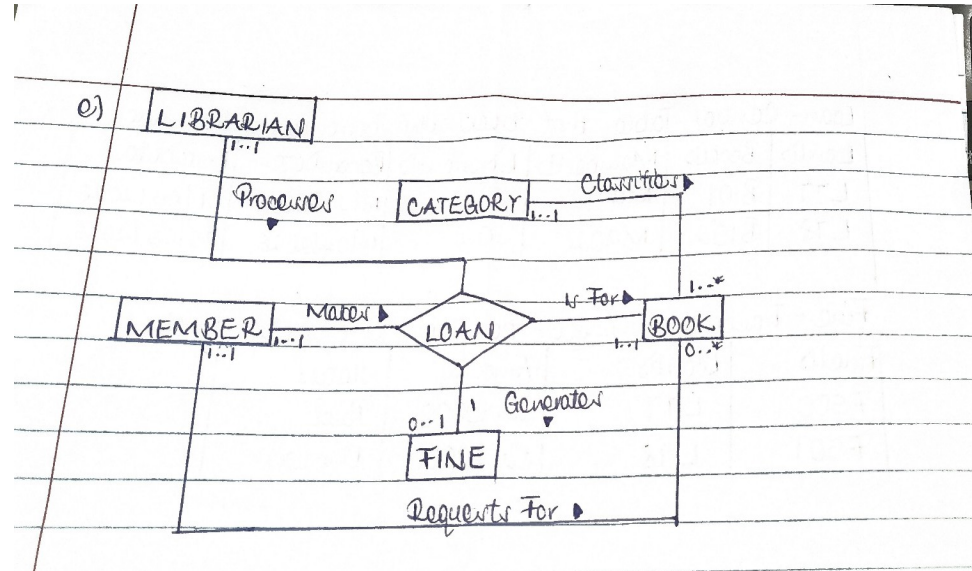


- Book and Category: Manages physical assets and subject classifications.
- Member and Librarian: Tracks authorized users and administrative staff.
- Loan: The central associative entity linking Members, Books, and Librarians.
- Fine: Manages financial penalties triggered by overdue items.

ENTITY RELATIONSHIP DIAGRAM (ERD)



- Logic: One Category contains many Books (1:N).
- Logic: One Member can have multiple Loan records (1:N).
- Logic: One specific Loan generates a maximum of one Fine (1:1).



PHYSICAL DATABASE DESIGN

IMPLEMENTATION IN MYSQL



- Schema: Translation of logical entities into interconnected tables using Foreign Keys.
- Data Integrity: Use of Primary Keys (PK) and Foreign Keys (FK) to link tables like Loan and Book.
- Automation: BorrowDate is automatically stamped to minimize manual entry errors.

SYSTEM TESTING AND QUERIES.



```
MySQL 8.0 Command Line Cli x + -
mysql> USE university_project;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_university_project |
+-----+
| book                          |
| category                     |
| fine                         |
| librarian                    |
| loan                         |
| member                       |
+-----+
6 rows in set (0.00 sec)

mysql> DESCRIBE loan;
+-----+
| Field      | Type | Null | Key | Default | Extra      |
+-----+
| LoanID     | int  | NO   | PRI | NULL    | auto_increment |
| BookID     | int  | YES  | MUL | NULL    |              |
| MemID      | int  | YES  | MUL | NULL    |              |
| LibID      | int  | YES  | MUL | NULL    |              |
| BorrowDate | date | YES  |     | NULL    |              |
| DueDate    | date | YES  |     | NULL    |              |
+-----+
6 rows in set (0.04 sec)

mysql> SELECT * FROM member;
+-----+
| MemID | FullName      | Email          |
+-----+
| 1     | John Aine     | john@univ.ac.ug |
| 2     | Jane Babirye | jane@univ.ac.ug |
| 3     | Musa Ali     | musa@univ.ac.ug |
+-----+
```



SYSTEM TESTING AND QUERIES.

- Validation: Sample data loaded for all six main areas to test relational links.
- Output: Successful execution of complex JOIN queries to identify active loans and responsible staff.

INDUSTRY BENCHMARKING

LEARNING FROM THE KOHA LMS



- Research: Studied the Koha Integrated Library System used globally.
- Application: Applied Koha's logic of separating structural data (Categories) from transactional data (Loans).
- Benefit: This prevents spelling errors and data anomalies during cataloging.

CONCLUSION AND PROJECT SUMMARY.



- Tangible Benefits: Reduced resource loss and automated fine calculations.
- Intangible Benefits: Improved data integrity and clear audit trails via Librarian IDs.
- Future Work: Expanding the system to support digital E-book downloads and remote reservations.